SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1707**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Geferlin Boffy R (192571080) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

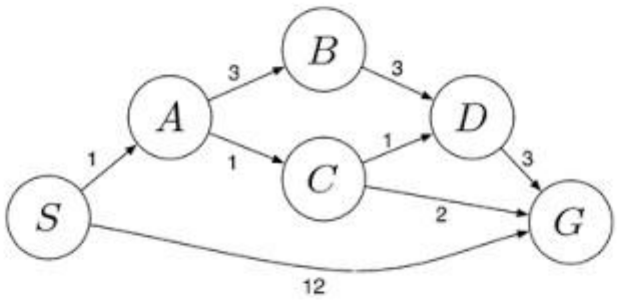
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			

ANSWERS:

1. Solve the Water Jug Problem
Problem Statement
Two jugs are available:

- One **4-gallon jug**
- One **3-gallon jug**

Both jugs have **no measuring marks**. Water can be:

- Filled from a pump,
- Emptied onto the ground,
- Poured from one jug to another.

Goal: Obtain **exactly 2 gallons** in the 4-gallon jug.

State Representation
State = (4-gallon jug, 3-gallon jug)
Initial State:
(0,0)
Goal State:
(2,0)

Solution Steps

Step	**State (4G,3G)**	**Action**
Initial	(0,0)	Both jugs empty
1	(0,3)	Fill the 3-gallon jug
2	(3,0)	Pour all water into the 4-gallon jug
3	(3,3)	Fill the 3-gallon jug again
4	(4,2)	Pour water into the 4-gallon jug until it becomes full
5	(0,2)	Empty the 4-gallon jug
6	(2,0)	Pour the remaining 2 gallons into the 4-gallon jug

Final Answer
The **4-gallon jug contains exactly 2 gallons**, which satisfies the goal.

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2. Mars Rover Agent
A Mars Rover is an autonomous intelligent agent designed to explore Mars, collect scientific data, and transmit information back to Earth.

i) What are the percepts of this agent?
Percepts are the information received from the environment through sensors.
The Mars Rover perceives:

- Camera images
- Soil and rock characteristics
- Temperature
- Atmospheric pressure
- Wind speed
- Sunlight intensity
- Terrain elevation
- Obstacles
- Battery level

- Position and direction
- Radiation levels

ii) Characterize the operating environment
The Mars Rover operates in the following AI environment:

Property	Description
Partially Observable	The rover cannot observe the complete environment at once.
Dynamic	Dust storms and weather conditions may change.
Sequential	Current actions affect future decisions.
Continuous	Movement, time and sensor readings are continuous.
Single Agent	Only one intelligent rover performs the mission.
Unknown	The terrain is not fully known before exploration.

iii) What actions can the agent perform?
The rover can:

- Move forward
- Move backward
- Turn left/right
- Capture images
- Collect soil samples
- Drill rocks
- Analyze samples
- Detect obstacles
- Avoid dangerous terrain
- Recharge through solar panels
- Send collected data to Earth

iv) How can the performance of the agent be evaluated?
Performance measures include:

- Accuracy of collected scientific data
- Number of successful sample collections
- Distance explored
- Efficient battery usage
- Safe obstacle avoidance
- Successful communication with Earth
- Completion of mission objectives

v) Suitable Agent Architecture
Utility-Based Learning Agent
Reason
A Utility-Based Learning Agent is the most suitable because:

- It selects actions that maximize mission success.
- It learns from previous experiences.
- It handles uncertain and changing environments.
- It manages energy efficiently.
- It makes intelligent decisions based on environmental conditions.

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3. Eight Queens Problem
Problem Statement
Place eight queens on an **8 × 8 chessboard** so that no queen attacks another queen.
A queen attacks horizontally, vertically and diagonally.

Problem Formulation

Initial State
An empty chessboard.

State Space
All possible arrangements of eight queens on the chessboard.

Operators

- Place a queen in a safe position.
- Move a queen to another row if a conflict occurs.

Goal Test
A solution is reached when:

- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are on the same diagonal.

Path Cost
Each move has equal cost.

Search Strategy
The most suitable search strategy is **Backtracking Search (Depth-First Search).**
Algorithm

1. Place the first queen in the first column.
2. Move to the next column.
3. Place the queen only if the position is safe.
4. If no safe position exists, backtrack.
5. Repeat until all eight queens are placed.

Advantages

- Uses less memory.
- Eliminates invalid solutions early.
- Guarantees a correct solution.
- Efficient for constraint satisfaction problems.

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4. OLA Cab Booking Problem
A customer wants to travel using the OLA mobile application by selecting different cab types such as Mini, Micro, Sedan, Prime or Shared.

Suitable Agent
Goal-Based Agent

Reason
A Goal-Based Agent is suitable because:

- Its goal is to transport the customer from the source to the destination.
- It evaluates available cab options.
- It selects the best cab according to customer preferences.
- It completes the booking only after achieving the desired goal.

Pseudo Code
START

Input source location
Input destination location

Display available cab types

Customer selects a cab

Check cab availability

IF cab is available THEN
    Calculate fare
    Confirm booking
    Assign nearest driver
    Start journey
    Complete trip
ELSE
    Display "Cab Not Available"
ENDIF

STOP

Working Process

1. Customer enters pickup location.
2. Customer enters destination.
3. System displays available cabs.
4. Customer selects a preferred cab.
5. Fare is calculated.
6. Booking is confirmed.
7. Driver is assigned.
8. Customer reaches destination.

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5. Uniform Cost Search (UCS)

Problem Statement

A logistics company wants to transport goods from warehouse **S** to warehouse **G** with the minimum transportation cost.
Uniform Cost Search expands the node with the **lowest cumulative path cost** first.

Graph Costs

Edge	**Cost**
S → A	1
S → G	12
A → B	3
A → C	1
B → D	3
C → D	1
C → G	2
D → G	3

Uniform Cost Search Execution

Step 1

Start at **S**
Frontier:
A (1)
G (12)
Expand **A** because it has the lowest cost.

Step 2
Expand **A**
New frontier:
C (2)
B (4)
G (12)
Expand **C**.

Step 3
Expand **C**
New frontier:
D (3)
G (4)
B (4)
Expand **D**.

Step 4
Expand **D**
New path:
G (6)
Existing path:
G (4)
Keep the lower cost path.

Step 5
Expand **G**.
Goal reached.

Optimal Path
S → A → C → G

Total Cost
S → A = 1
A → C = 1
C → G = 2

Total Cost = 4

Search Order
S
↓
A
↓
C
↓
D
↓
G

Why UCS Gives the Optimal Solution

- Expands nodes based on the **lowest cumulative cost**.
- Guarantees the **minimum-cost path** when all edge costs are non-negative.
- Does not stop until the least-cost goal node is reached.